

Connor W. Magoon

Education	The University of North Carolina at Chapel Hill (UNC) PhD Applied Mathematics 2023–2028 Thesis advisor: Pedro Sáenz, Secondary advisor: Shahar Kovalsky BS Physics and Applied Mathematics 2019–2023
Positions	Simons Foundation Jun–Aug 2026 Summer Pre-Doctoral Researcher Flatiron Institute / Center for Computational Biology / Biophysical Modeling Group Mentors: Scott Weady, David Stein
Publications	Differentiation Through Black-Box Quadratic Programming Solvers 2025 C. W. Magoon*, F. Yang*, N. Aigerman, S. Z. Kovalsky <i>NeurIPS</i> Code Poster The way bubbles gallop 2025 J. H. Guan, S. I. Tamim, C. W. Magoon, H. A. Stone, and P. J. Sáenz <i>Physical Review Fluids</i> Video Galloping Bubbles 2025 J. H. Guan*, S. I. Tamim*, C. W. Magoon*, H. A. Stone, and P. J. Sáenz <i>Nature Communications</i> Traveling Faraday waves 2023 J. H. Guan, C. W. Magoon, M. Durey, R. Camassa, and P. J. Sáenz <i>Physical Review Fluids</i> Video *Equal contribution
Submitted	Spontaneous galloping of a vibrating bubble along a solid boundary. Part 1: Experiments J. H. Guan, X. Liu, S. I. Tamim, C. W. Magoon, and P. J. Sáenz Spontaneous galloping of a vibrating bubble along a solid boundary. Part 2: Simulations C. W. Magoon, S. I. Tamim, and P. J. Sáenz Spontaneous galloping of a vibrating bubble along a solid boundary. Part 3: Theory S. I. Tamim, C. W. Magoon, and P. J. Sáenz
Awards	<i>Travel Grants</i> NeurIPS Scholar Award 2025 <i>Conference Awards</i> Gallery of Fluid Motion Award 2024 J. H. Guan, S. I. Tamim, C. W. Magoon, H. A. Stone, and P. J. Sáenz, “Galloping Bubbles” <i>American Physical Society / Division of Fluid Dynamics</i> Milton van Dyke Award 2022 J. H. Guan, C. W. Magoon, M. Durey, R. Camassa, and P. J. Sáenz, “Run, Faraday, Run” <i>American Physical Society / Division of Fluid Dynamics</i> <i>Honors</i> Doctoral Merit Fellowship, UNC 2023–2024 Archibald Henderson Mathematics Medal, UNC 2023

Talks	Collective Dynamics of Galloping Bubbles	2025
	C. W. Magoon, X. Liu, S. I. Tamim, H. A. Stone and P. J. Sáenz APS Division of Fluid Dynamics Meeting	
	Collective Dynamics of Galloping Bubbles	2024
	C. W. Magoon, X. Liu, J. H. Guan, S. I. Tamim, and P. J. Sáenz APS Division of Fluid Dynamics Meeting	
	Direct Numerical Simulations of Vibrating Bubbles Self-Propelling Along Solid Boundaries	2023
	C. W. Magoon, J. H. Guan, S. I. Tamim, and P. J. Sáenz APS Division of Fluid Dynamics Meeting	
Teaching	<i>Instructor</i>	
	Differential Equations Lab (UNC MATH 383L)	Fall 2025
	<i>Teaching Assistant</i>	
	Multivariable Calculus (UNC MATH 233)	Fall 2024, 2025
	General Physics (UNC PHYS 115)	Summer 2021
	Introductory Electromagnetism (UNC PHYS 119)	Fall 2020
	<i>Grader</i>	
	Fluid Mechanics (UNC MATH 563)	Spring 2025
	Differential Equations (UNC MATH 524)	Spring 2024